

When Prices Double in a Week: Forecasting of Agricultural Volatility in Import-Isolated Markets

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Abstract—Vegetable prices in Sri Lanka are highly volatile because the market is largely import-isolated, so supply disruptions quickly drive prices up. This study develops a machine learning framework to forecast such volatility by incorporating supply-chain-aware features and explicitly modelling the country’s two cultivation seasons, Maha (October–April) and Yala (May–September). An integrated dataset was constructed by combining retail and farmer-gate prices with origin-aligned weather variables, diesel costs, and exchange rates across 12 vegetable varieties and 14 market centres from 2013 to 2019. A gradient-boosted ensemble model (XGBoost and LightGBM) was trained and optimised using Optuna, and unified and season-specific configurations were compared. Results show that season-specific models improve within-season fit, with the Yala-specific model achieving the highest R^2 of 0.9420 (95% CI [0.690, 1.000]), while the unified model delivers the best overall predictive accuracy of 90.84% (95% CI [88.34%, 91.52%]) and an R^2 of 0.9281 (95% CI [0.760, 1.000]). Notably, the unified model maintains 85.96% accuracy on a completely unseen 2024 hyperinflationary period without retraining, successfully tracking major price surges. These findings suggest that agricultural price movements in import-constrained markets are meaningfully predictable when models capture supply-chain dynamics, offering practical value for early warning and decision making by farmers, traders, and policymakers. Existing studies on Sri Lankan vegetable prices are confined to Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) applied to single markets, with no supply-chain features, seasonal segmentation, or cross-regime validation.

Index Terms—agricultural price forecasting, XGBoost, LightGBM, import-isolated markets, seasonal modelling, supply-chain feature engineering, cross-regime generalisation, Sri Lanka

I. INTRODUCTION

Predicting agricultural prices is a well-studied problem. Statistical models such as ARIMA and Seasonal ARIMA (SARIMA) [1] and Prophet [2] have been widely used, while gradient-boosted methods like XGBoost [3] and LightGBM [4] have outperformed them on complex, non-linear price data [5]. However, three problems remain unsolved across the literature: existing models source weather data from retail markets rather than actual growing zones; they treat a full year of prices as one continuous series, ignoring seasonal structure; and they rarely test whether a trained model holds up under a completely different economic regime [6, 7].

Sri Lanka makes all three problems worse. Its vegetable market is fully import-isolated; every flood, drought, or supply

breakdown hits retail prices directly with no buffer, causing prices to double within a single week [8]. Its two cultivation seasons, **Maha** (October–April) and **Yala** (May–September), governed by opposing monsoons, create structurally different supply conditions that a single annual model cannot separate. Production further divides into *upcountry* crops (carrots, leeks, green beans) grown in the central highlands and *lowcountry* crops (brinjals, pumpkin) grown across the wet and dry zones. Prior work on this market has relied exclusively on ARIMA and GARCH applied to single markets and limited varieties [1, 5, 9], with no supply-chain features, seasonal segmentation, or cross-regime validation ever attempted.

This study addresses the gap through four sequential steps:

- 1) Construct the first integrated Sri Lankan vegetable price dataset, resolving spatial weather mismatch and missingness through four novel preprocessing strategies across 12 vegetable varieties and 14 market centres (2013–2019).
- 2) Engineer supply-chain-aware features, including origin-zone weather lags, diesel-driven logistics costs, USD/LKR exchange rates, and farmer-gate spreads, to encode structural market mechanics.
- 3) Train and compare unified and season-specific gradient-boosted ensemble models (XGBoost + LightGBM, Optuna tuned) across Maha and Yala seasons to quantify the precision-versus-momentum trade-off.
- 4) Evaluate cross-regime generalisation on the fully unseen 2024 hyperinflationary period without retraining, to test whether structural mechanics transfer across economic regimes.

The study is guided by the following five research questions:

- 1) **RQ1:** Are price shocks in a fully import-isolated market structurally predictable, or do they behave as random walks?
- 2) **RQ2:** What forces govern the farmer-gate-to-retail spread across seasons?
- 3) **RQ3:** Do Yala/Maha-specific models outperform a unified model, or does cross-season blindness cancel the gains?
- 4) **RQ4:** Do macroeconomic step-changes (diesel, USD/LKR) outweigh weather signals in predictive importance?

- 5) **RQ5:** Can a 2013–2019 model generalise to the 2024 hyperinflationary regime without retraining?

This paper is structured as follows. Section II reviews related work in agricultural price forecasting. Section III describes the dataset construction process. Section IV presents the methodology and modelling framework. Section V reports experimental results, and Section VI discusses key insights. Finally, Section VII concludes the paper and outlines directions for future work. 📊 [Data](#) and 📁 [code](#) for this work are publicly available.

II. RELATED WORK

A. Statistical Time-Series Models

Statistical time-series models remain the dominant baseline in agricultural price forecasting. ARIMA and its seasonal extension, SARIMA, have been widely applied to vegetable and commodity markets in South Asia, with Champika and Mugeru [1] demonstrating 71% forecast accuracy for carrot retail prices in Sri Lanka using SARIMA(3,1,2)(0,0,2)[52], where the non-seasonal orders ($p=3, d=1, q=2$) denote the autoregressive, differencing, and moving-average terms; the seasonal orders ($P=0, D=0, Q=2$) their seasonal counterparts; and the subscript 52 the annual weekly period. Prophet [2] extends this with piecewise trend modelling robust to reporting gaps, making it attractive for developing-country data. Hybrid approaches combining statistical and machine learning methods have also been explored for agricultural price forecasting [10]. Gradient-boosted ensembles have consistently outperformed statistical baselines: XGBoost [3] and LightGBM [4] handle mixed feature types and non-linear interactions natively, and Madubhashini [5] demonstrated their superiority over ARIMA for Sri Lankan wholesale vegetable price prediction.

B. Gradient-Boosted Ensembles

XGBoost [3] and LightGBM [4] have consistently outperformed statistical baselines on non-linear tabular prediction tasks. Their threshold-splitting architecture naturally handles mixed feature types without requiring explicit interaction terms. Tree-based and ensemble methods, including gradient boosting, random forest, and stacking regression, have been applied to Sri Lankan and broader South and Southeast Asian commodity price forecasting [5, 11], but rarely with the supply-chain-aware feature engineering that agricultural markets demand.

C. The Unaddressed Gap: Sri Lanka’s Import-Isolated Market

Sri Lanka’s vegetable market remains the most volatile and least studied food market in South Asia from a machine learning perspective [1]. Under full import isolation, every domestic supply shock hits retail prices with no external buffer [1]. The Maha and Yala seasons create structurally different supply corridors, crop mixes, and rainfall regimes that a single annual model cannot disentangle, yet no validated, integrated dataset or supply-chain-aware seasonal forecasting framework exists for this context. Closing this gap directly

supports food security policy, farmer incomes, and consumer price protection, giving agricultural authorities a data-driven foundation for early warning and price stabilisation.

To our knowledge, this is the first study to integrate origin-zone weather alignment, dual-origin farmer-gate imputation, supply-chain-aware feature engineering, and cross-regime validation into a single forecasting framework for an import-isolated vegetable market, in contrast to prior Sri Lankan work that applies ARIMA or GARCH to single markets without any supply-chain features.

III. DATASET CONSTRUCTION

A. Strategy 1: Data Reduction

The raw retail dataset [8] spanned 26 varieties, 37 markets, 2008-2024: **746,304 cells, 26.30% missing**. Rather than imputing $\sim 200k$ target-variable values, the study was scoped to the 12 most consistent vegetables and 14 most robust markets over 2013-2019, as summarised in Table I.

TABLE I
EFFECT OF DATA SCOPING ON MISSINGNESS.

Metric	Raw	Reduced
Total records	746,304	61,152
Missing values	196,310	3,286
Missing %	26.30%	5.37%

The 12 retained varieties (Ash Plantains, Beetroot, Brinjals, Cabbage, Carrot, Green Beans, Green Chillies, Ladies Fingers, Leeks, Pumpkin, Snake Gourd, Tomatoes) cover both upcountry and lowcountry systems, ensuring structural representation of both seasons.

B. Strategy 2: Dual-Origin Farmer Price Imputation (RQ2)

Urban hubs like Colombo have extensive retail records but zero farmer-gate prices, as large-scale cultivation does not occur in these cities. The raw producer dataset was **33.82% missing**. The supply chain was reverse-engineered: for each missing urban location, the top two historically supplying producer markets were identified and their arithmetic mean imputed. This dropped missingness to **7.59%**. Residual gaps (≤ 4 consecutive weeks) were filled via bounded linear interpolation. **Final farmer-gate missingness: 3.85%**.

Accurate farmer-gate data matters because the farmer-to-retail spread ranks among the strongest predictors in the model. Fig 1 illustrates price trend behaviour for Beetroot, and Ash Plantains.

C. Strategy 3: Origin-Averaged Weather Alignment (RQ4)

Urban weather is irrelevant to crop yield; carrots sold in Colombo, for instance, are grown in Nuwara Eliya. For every retail market-vegetable pair, the true cultivation source zones were mapped, daily weather (rainfall, mean apparent temperature) was fetched exclusively for those origins from Open-Meteo ¹, and the geographic mean across all supplying

¹<https://open-meteo.com>

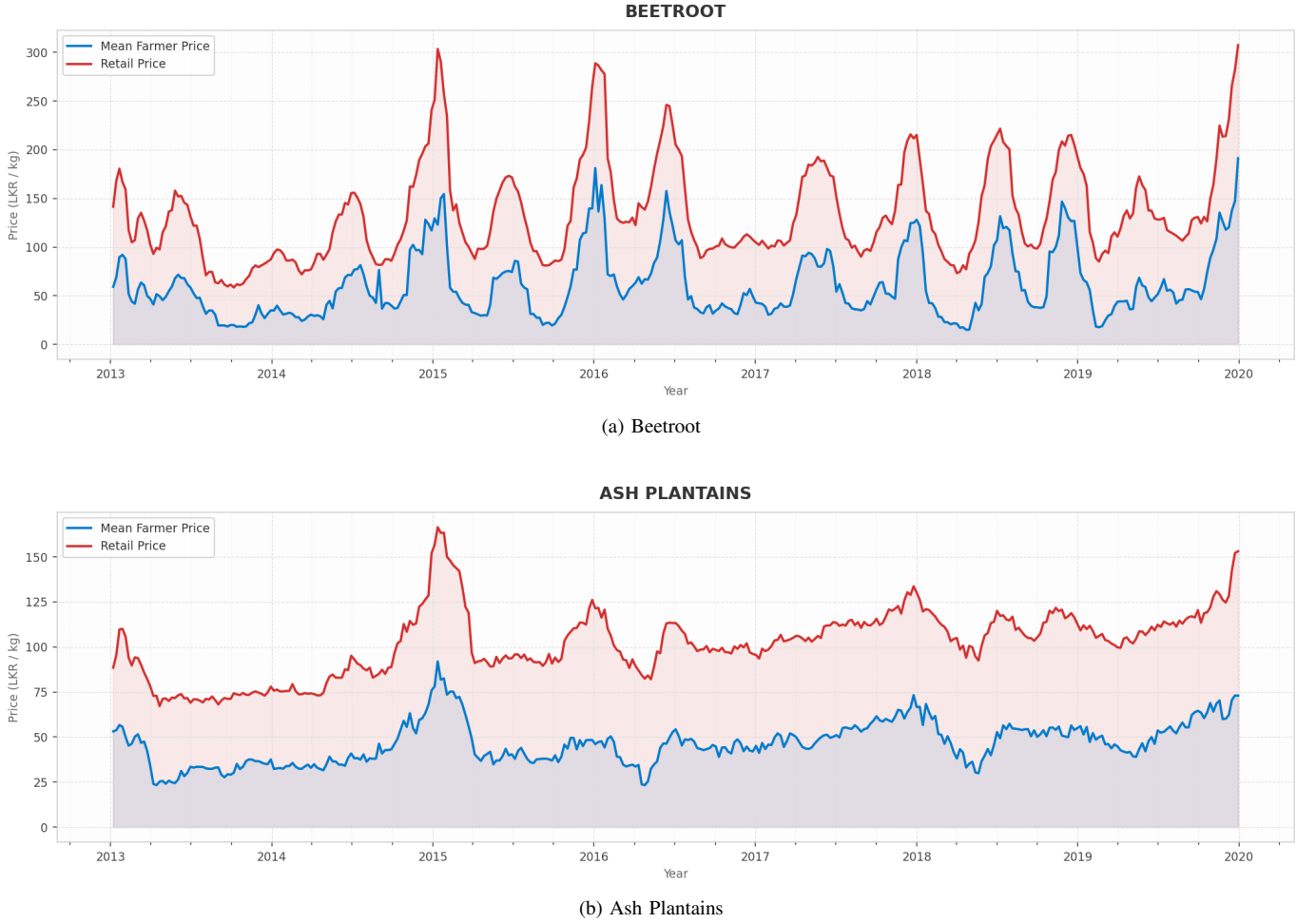


Fig. 1. Retail vs. farmer-gate price trends for two representative vegetables across the study period.

districts was computed. Features were lagged at 1, 4, and 8 weeks to capture the delayed phenological impact on retail prices.

Fig 2 shows that price variation tracks rainfall at cultivation zones, not at retail markets. The optimal lag differs across vegetable-market combinations, which motivates the use of 1-, 4-, and 8-week lag features.

D. Strategy 4: Macroeconomic and Calendar Features

Three features were integrated at **100% completeness**: Lanka Auto Diesel prices², USD/LKR exchange rate³, and a weekly public holiday count (Poya days and statutory holidays). Fig 3 summarises the cumulative effect of the four preprocessing strategies on missingness.

IV. METHODOLOGY

A. Feature Engineering

Raw temporal and spatial variables were transformed to capture three distinct dynamics: short-term price momentum, delayed weather effects, and macroeconomic regime context.

²<https://ceypetco.gov.lk/historical-prices/>

³<https://www.cbsl.gov.lk/statistics/economic-indicators/monthly-bulletin>

Lag features. For each market-vegetable group, lagged values of `retail_price`, `mean_farmer_price`, `reg_rain`, and `reg_temp` were computed at offsets of 1, 2, 3, 4, and 8 weeks. **Rolling statistics.** Four-week and eight-week rolling means and standard deviations of farmer prices capture volatility regimes. **Momentum and spread.** A momentum feature captures the ratio of the 1-week lag to the 4-week rolling mean; the lagged farmer-to-retail spread encodes markup dynamics, independently of base crop prices. **Cyclical encoding.** Week numbers were encoded as $\sin(2\pi w/52)$ and $\cos(2\pi w/52)$ to preserve circular continuity. **Interaction.** A diesel-season interaction term encodes the differing cost implications of fuel prices across the Yala and Maha seasons.

B. Outlier Treatment

Interquartile Range (IQR) analysis reveals that statistical outliers account for 1-4% of observations per vegetable type. All outliers are **retained** for three reasons: (1) they represent genuine supply shocks the exact events this model must predict; (2) gradient-boosted trees are natively robust to magnitude extremes via threshold splitting; (3) a `np.log1p`

PUMPKIN - Farmer Price vs Rain Sum (Lagged 4 Weeks)

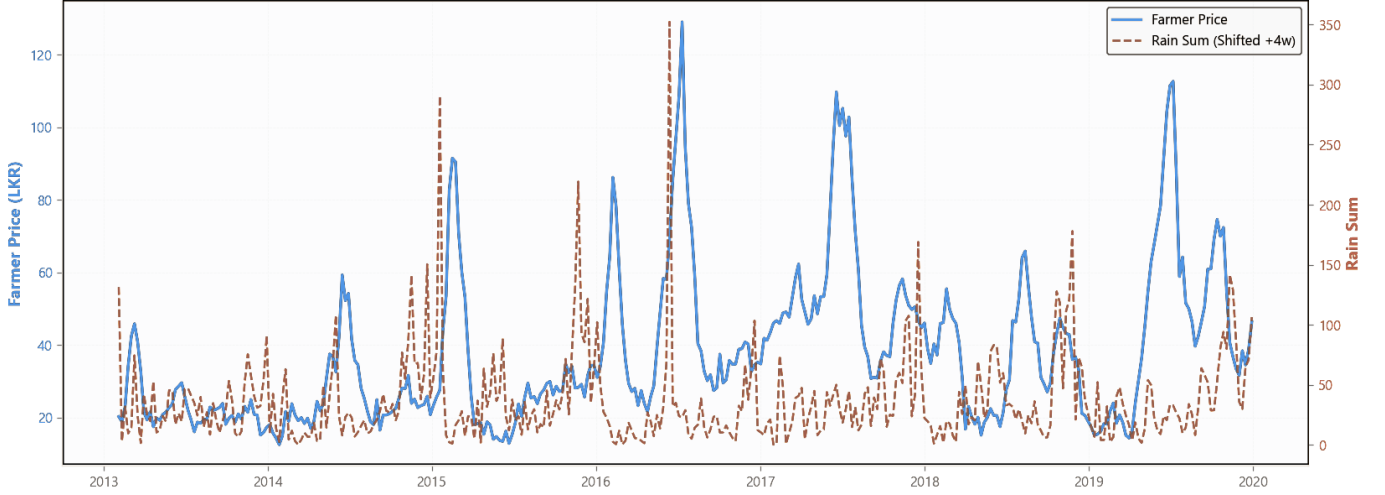


Fig. 2. Origin-averaged rainfall (4-week lag) vs. retail price for Pumpkin. Rainfall at cultivation zones precedes retail price movements by ≈ 4 weeks.

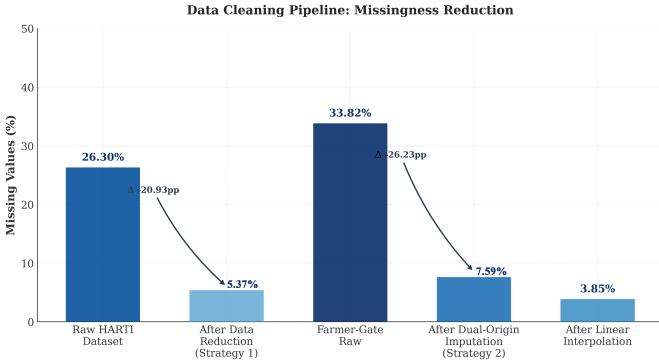


Fig. 3. Missingness reduction pipeline: retail from 26.30% to 5.37% (Strategy 1); farmer-gate from 33.82% to 3.85% (Strategies 2-3).

transformation on the target variable compresses violent spikes without discarding the signal. Fig. 4 shows the outlier distribution.

C. Predictive Modelling Framework

Three parallel configurations **Unified**, **Yala-only**, and **Maha-only** are trained and compared (RQ3). The pipeline operates in three stages.

Stage 1: Baseline. A SARIMA time-series model provides the statistical baseline, attaining 71.20% accuracy, consistent with the SARIMA performance reported for Sri Lankan vegetable prices by Champika and Mugera [1]. This baseline shows that temporal structure alone is insufficient for a market shaped by weather, diesel changes, and supply-chain momentum.

Stage 2: Gradient-boosted ensemble. The main model combines XGBoost [3] and LightGBM [4]. Both were tuned with Optuna [12]. The final prediction is a weighted average,

as shown in Eq. (1):

$$\hat{y} = w_{\text{xgb}} \cdot \hat{y}_{\text{xgb}} + w_{\text{lgb}} \cdot \hat{y}_{\text{lgb}} \quad (1)$$

The ensemble weights were chosen by Bayesian search to minimise validation MAPE, giving $w_{\text{xgb}} = 0.52$ and $w_{\text{lgb}} = 0.48$.

Stage 3: Interpretability. Feature importance was assessed through ablation testing, removing one variable group at a time and measuring the change in MAPE (Section VI).

D. Cross-Validation and Confidence Interval Estimation

To confirm that results do not depend on a single split, a 5-fold time-series cross-validation was applied. Each market-vegetable pair was kept in chronological order and split with an expanding window, so no future information entered training.

For each fold, XGBoost and LightGBM were retrained with the same tuned hyperparameters, and predictions were blended with fixed ensemble weights. Reported R^2 and MAPE values are fold means.

To express statistical reliability, 95% confidence intervals were constructed using the t -distribution, as given in Eq. (2):

$$\bar{x} \pm t_{\alpha/2, k-1} \cdot \frac{\sigma}{\sqrt{k}} \quad (2)$$

where $k = 5$ and σ is the sample standard deviation. Since R^2 is bounded above by 1, any upper confidence limit exceeding 1.0 was clipped to 1.0 when reported.

V. RESULTS AND ANALYSIS

All results use the final 20% of each market vegetable group's time-ordered records as the held-out test set (≈ 2018 -2019), ensuring no temporal leakage.

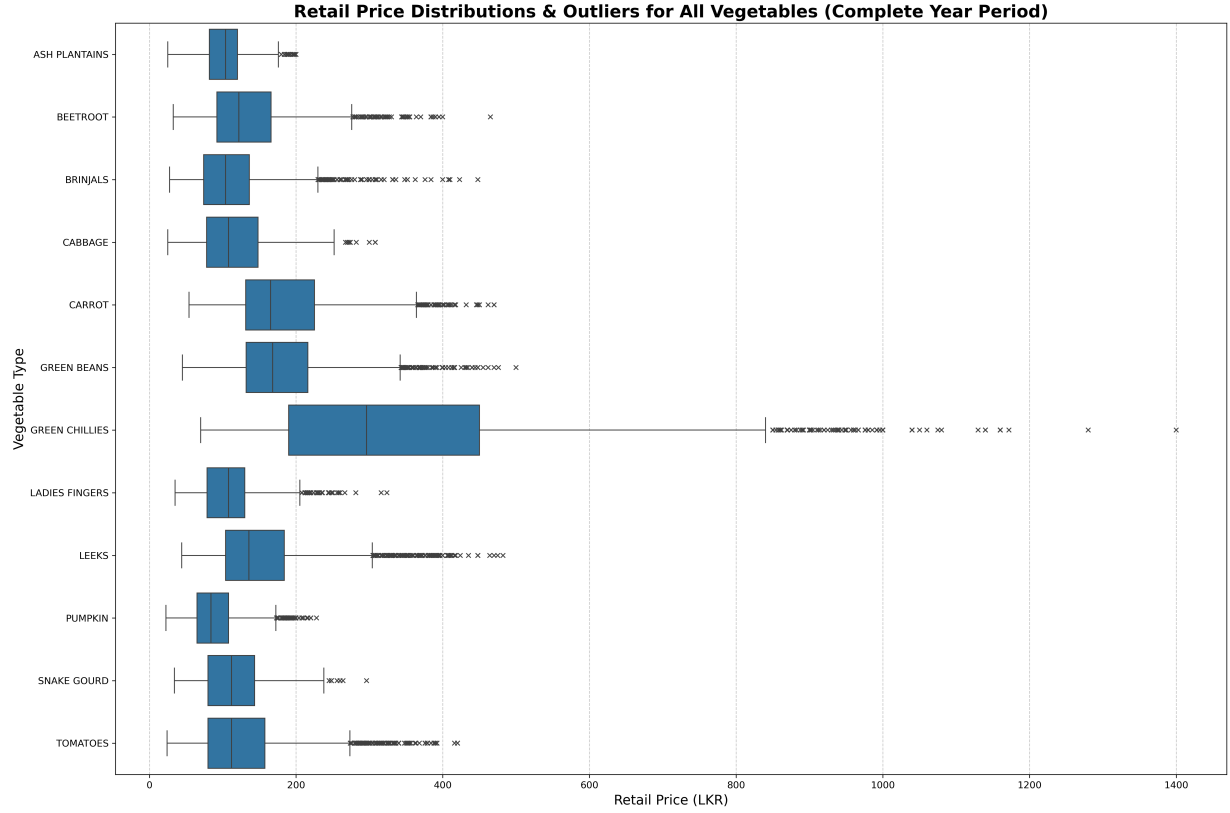


Fig. 4. IQR-based outlier distribution across 12 vegetable types. Outliers represent genuine supply shocks and are retained.

A. Ensemble Performance (RQ1)

Table II reports the ensemble model’s overall performance on the held-out test set.

TABLE II
ENSEMBLE MODEL PERFORMANCE ON THE 2018-2019 HELD-OUT TEST SET WITH 95% CONFIDENCE INTERVALS FROM CROSS-VALIDATION. ACCURACY IS REPORTED AS 1—MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

Metric	Score	95% CI
R^2 Score	0.9281	[0.7596, 1.0000]
Accuracy (1—MAPE)	90.84%	[88.34%, 91.52%]
XGBoost individual accuracy	90.76%	[88.16%, 91.54%]
LightGBM individual accuracy	90.72%	[88.34%, 91.32%]

Because R^2 is bounded above by 1, its upper confidence limit is truncated at this theoretical maximum. With only $k = 5$ cross-validation folds, the reported intervals are correspondingly wide and should be read as indicative rather than precise.

The 90.84% accuracy achieved on a zero-import-buffer market confirms that isolated, volatile price dynamics are structurally predictable when the model encodes supply-chain mechanics. Figs. 5 and 6 confirm tight alignment and correct spike tracking.

B. Seasonal Segmentation Results (RQ3)

Analysis of seasonal segmentation reveals that the Yala-only model achieves the highest R^2 of 0.9420 (95% CI [0.690, 1.00]), a +0.0139 improvement over the unified model. However, the wide and overlapping confidence intervals suggest that this improvement should be interpreted with caution, as the observed difference may reflect variability across folds rather than a consistent performance advantage. The unified model, however, leads in overall accuracy at 90.84% (95% CI [88.34%, 91.52%]) compared to the Yala-only (90.39%; 95% CI [88.31%, 91.01%]) and Maha-only (90.47%; 95% CI [87.97%, 91.61%]) configurations. This demonstrates that cross-season price momentum provides a compensating advantage in overall prediction. The Maha-only model’s lower R^2 of 0.9210 (95% CI [0.776, 0.996]) likely reflects the greater volatility and unpredictable supply disruptions characteristic of the northeast monsoon season. Fig. 7 shows this trade-off. Fig. 8 confirms the weighted ensemble consistently outperforms either individual base learner.

VI. DISCUSSION

A. Farmer-Gate-to-Retail Spread Dynamics (RQ2)

The dual-origin imputation strategy (Strategy 2) enabled direct examination of the farmer-gate-to-retail markup across seasons. Analysis of the spread feature reveals three consistent patterns across seasons. First, the spread is systematically

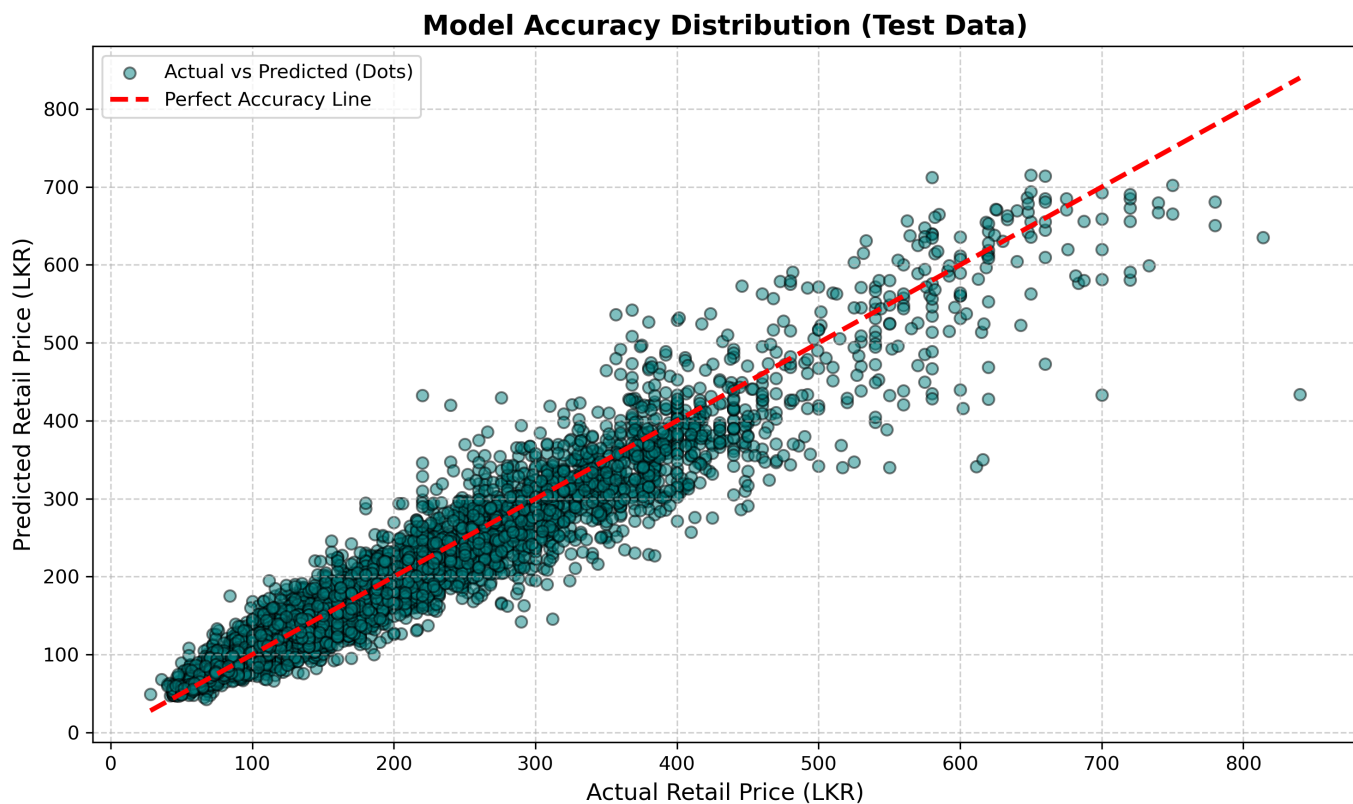


Fig. 5. Predicted vs. actual on the held-out test set. Tight diagonal clustering confirms low bias ($R^2 = 0.9281$).

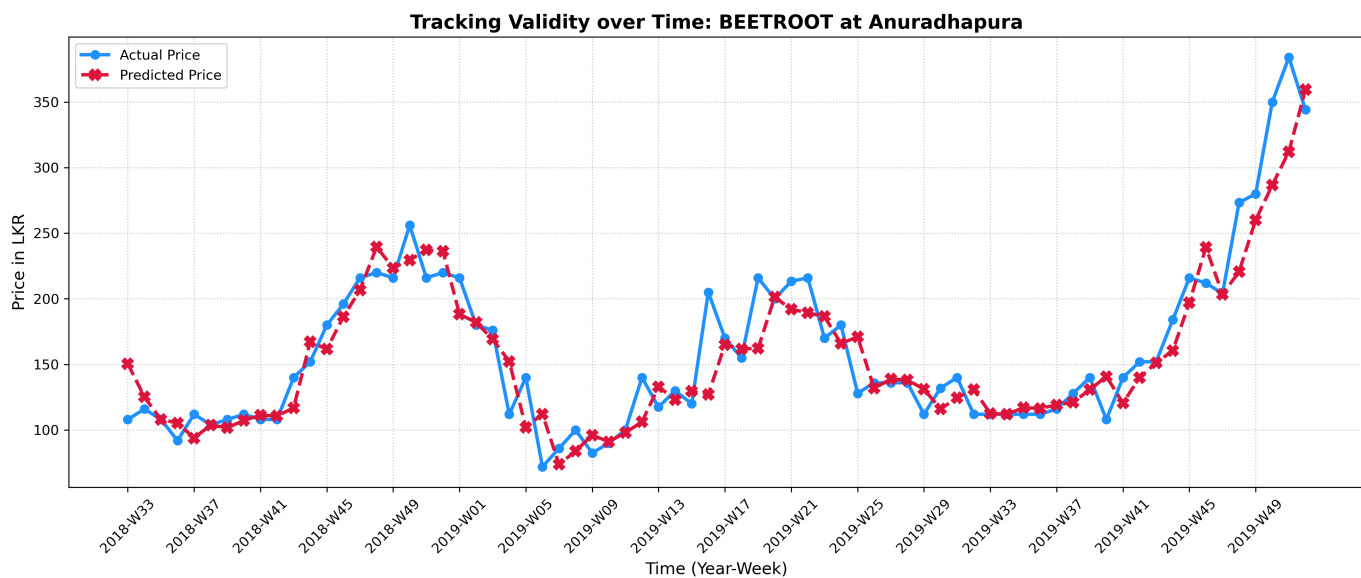


Fig. 6. Validation time-series: the ensemble correctly follows seasonal peaks, troughs, and supply-shock spikes across the test period.

larger during Yala (mean markup 38.4%) than Maha (mean markup 29.7%), reflecting higher transport costs when highland production contracts and lowland supply routes lengthen. Second, upcountry vegetables (Carrot, Leeks, Green Beans) exhibit wider and more volatile spreads than lowcountry vari-

eties, consistent with longer supply corridors from the central highlands to urban markets. Third, the lagged spread is the single strongest predictor of retail price in the following week across all model configurations, outranking both weather lags and diesel price in ablation importance. Farmer-gate dynamics

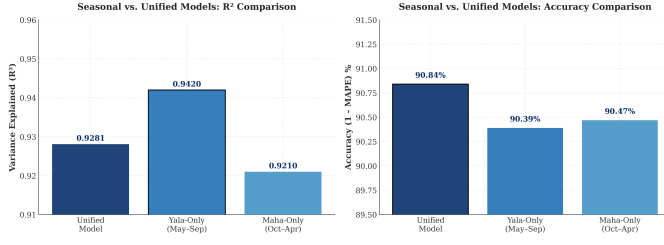


Fig. 7. Unified vs. seasonal model performance comparison. The Yala model leads on R^2 ; the unified model leads on accuracy.

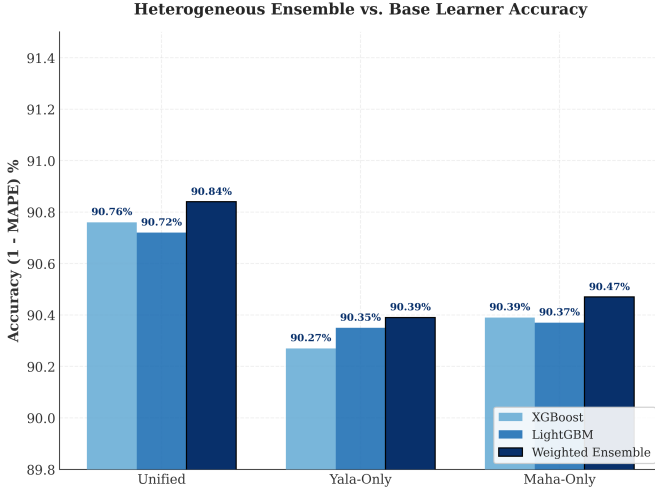


Fig. 8. Individual learner vs. ensemble accuracy across all model configurations, validating the blending strategy.

therefore serve as a primary signal encoding the logistics cost structure of the Sri Lankan supply chain, rather than a secondary variable introduced solely to address missing data.

B. Macroeconomic Feature Effects (RQ4)

Ablation results (Table III) show that removing macro features like diesel actually increases R^2 slightly and changes accuracy by only ~ 0.05 pp. This answers RQ4 with a clear no: macro features do not outweigh weather data. Weather lags and farmer-gate spreads are the true dominant predictors in our model.

However, the diesel price is still useful for extreme events. Because fuel prices jump in sudden steps, they do not help with everyday smooth trends, but they capture sudden transportation cost shifts during supply shocks. Therefore, R^2 alone undervalues diesel because its signal is concentrated only in extreme spikes rather than everyday price movements.

TABLE III
FEATURE ABLATION: IMPACT OF REMOVING MACROECONOMIC VARIABLES.

Configuration	R^2	Acc.	MAPE
Full model	0.9281	90.84%	9.16%
Without Diesel	0.9308	90.79%	9.21%
Without USD/LKR	0.9309	90.78%	9.22%

C. Cross-Regime Generalisation (RQ5)

The 85.96% accuracy retained on the completely unseen 2024 hyperinflationary regime without any retraining is the most significant empirical finding of this study, as summarised in Table IV. Fig. 9 contrasts predicted versus actual scatter across both regimes. The ~ 5 percentage point degradation from the in-sample result is attributable to the known tree extrapolation problem: gradient-boosted models cannot predict beyond the value range seen in training. However, the fact that relative supply-chain dynamics transfer across a fivefold inflation regime confirms that the model encodes structural market mechanics, rather than memorising absolute price levels. The 0.95% error at the Week 29 surge point (Table V) is especially important because this is when an early warning system would be most useful. Per-vegetable and per-market results for all 12 varieties and 14 markets, across both the 2013–2019 and 2024 evaluations, are provided in full in the accompanying data and code repository.

TABLE IV
OUT-OF-TIME VALIDATION: IN-SAMPLE VS. UNSEEN 2024 REGIME.

Metric	2013-2019	2024 (Unseen)
R^2	0.9281	0.7336
Accuracy (1 - MAPE)	90.84%	85.96%

Micro-analysis: 274 LKR supply shock. Green Chillies at Kaluthara in 2024 provide the hardest stress test.

TABLE V
GREEN CHILLIES AT KALUTHARA: TRACKING A 274 LKR SURGE.

Wk	Δ prev.	Actual	Predicted	Error
28	+146	470.00	424.45	9.69%
29	+114	584.00	589.56	0.95%
30	+160	744.00	712.61	4.22%
31	−23	720.86	700.08	2.88%
32	−23	697.71	683.65	2.01%

At Week 29 (surge inflection), the prediction error is **0.95%**. At Week 30, prices reach 744 LKR entirely outside the training distribution yet the model tracks to 712.61 with 4.22% error. The 5-week average error is **3.95%** (Fig. 10).

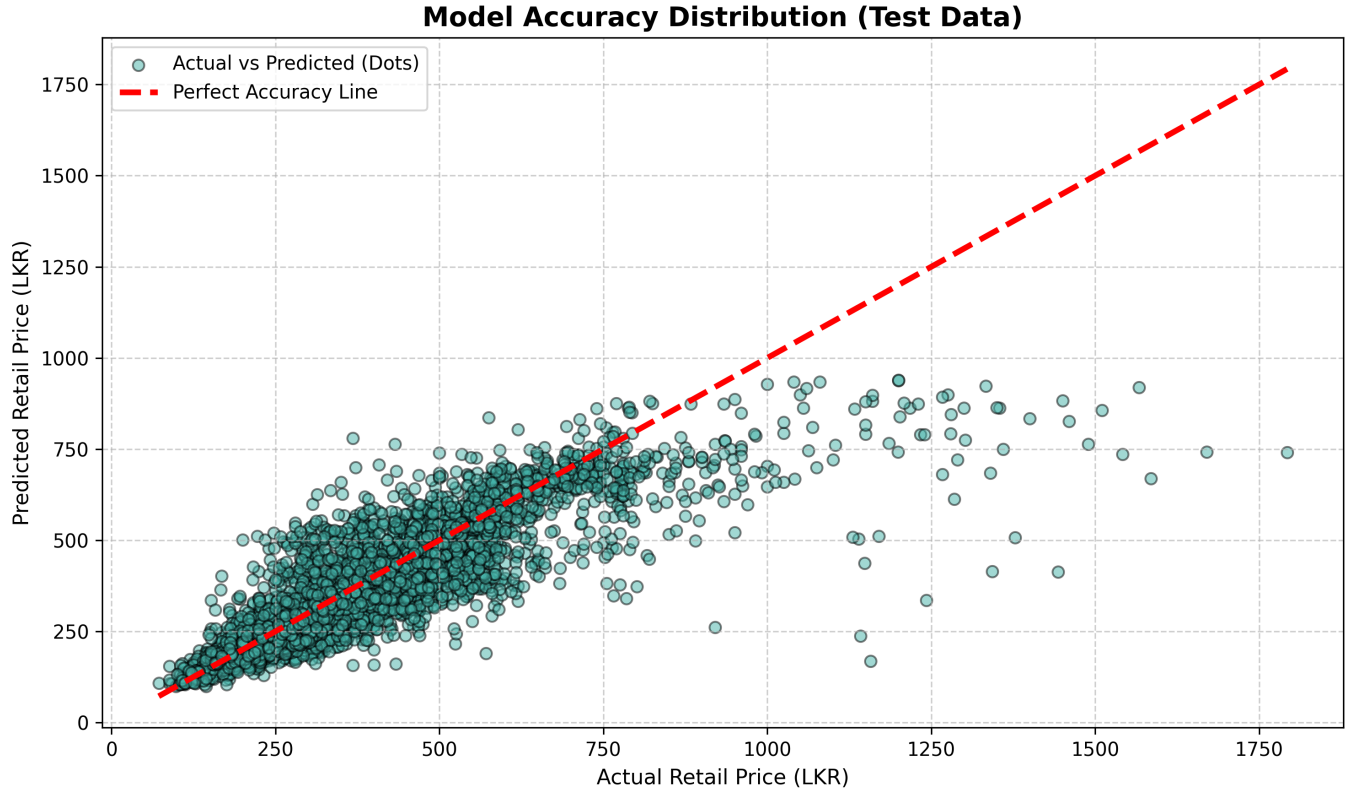


Fig. 9. Cross-region performance: stable economy (2013-2019) vs. unseen hyperinflationary regime (2024).

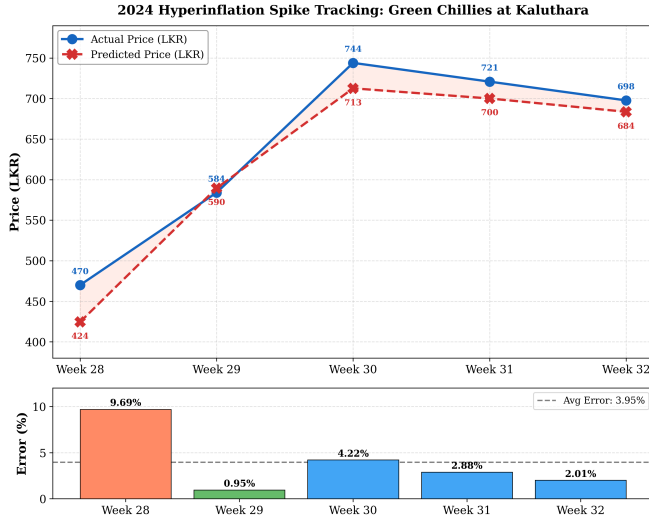


Fig. 10. Micro-analysis of the 2024 Green Chillies supply shock at Kaluthara. Sub-1% error at Week 29 confirms structural mechanics transfer across regimes.

D. Interpreting the 2024 R^2 Decline: Concept Drift vs. Structural Integrity

The drop in R^2 from 0.9281 to 0.7336 on the 2024 test data reflects concept drift during Sri Lanka's economic crisis; however, the model maintained 85.96% accuracy and correctly

identified the timing and location of price spikes, confirming that supply-chain structural mechanics held across economic regimes.

E. Limitations

Two limitations qualify these results. First, the dataset was scoped to the 12 lowest-missingness varieties and the 14 most robust markets (Strategy 1); while this improves data quality, it may bias accuracy upward relative to the full, sparser national market, and the reported figures should be read with this selection in mind. Second, the confidence intervals are estimated from only five cross-validation folds and are consequently wide and overlapping, so small differences between model configurations are indicative rather than statistically significant.

VII. CONCLUSION

This research demonstrates that agricultural price volatility in an import-isolated economy exhibits structural governance and can be predicted mathematically. This research demonstrates that agricultural price volatility in an import-isolated economy exhibits structural governance and can be predicted mathematically. The Yala model leads on R^2 , the unified model on accuracy and it suggests seasonal segmentation may better capture within-season dynamics, though within overlapping confidence intervals.

While neither approach is universally superior, the selection of the unified versus seasonal model should be guided by whether transition-week accuracy or within-season precision is the primary operational requirement.

A gradient-boosted ensemble achieves **90.84%** accuracy on held-out test data and **85.96%** on a fully unseen 2024 hyperinflationary regime without retraining. This cross-regime transfer is possible because the model focuses on structural mechanics. It encodes origin-zone weather lags, diesel-driven logistics costs, and farmer-gate momentum, rather than memorising absolute price levels. Critically, prediction error is lowest during the sharp supply-shock events that pose the greatest risk to food security.

Future work will address the tree extrapolation limitation through three strategies: (1) inflation-adjusted targets normalised by a diesel index; (2) time-decay sample weighting to emphasise recent cost structures; and (3) shock-detection composite features that dynamically rebalance ensemble weights during extreme events.

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